TWO BASE-CHANGE STABLE PROPERTIES REALISED BY APPROXIMATIONS CAN BE REALISED SIMULTANEOUSLY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10001-10034 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10036-10072, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian approximation of finitely presented morphisms of affine schemes, base-change stability of the two properties, and finite type bases over Z; the combination is retained.

Source reference. Original lemma 01ZM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10072–10072. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *(Noetherian approximation and combining properties.) Let P, Q be properties of morphisms of schemes which are stable under base change. Let $f : X \to S$ be a morphism of finite presentation of affine schemes. Assume we can find cartesian diagrams*

$$\begin{array}{ccc} X_1 & \longleftarrow & X \\ {\scriptstyle f_1}\downarrow & & \downarrow{\scriptstyle f} \\ S_1 & \longleftarrow & S \end{array} \quad \text{and} \quad \begin{array}{ccc} X_2 & \longleftarrow & X \\ {\scriptstyle f_2}\downarrow & & \downarrow{\scriptstyle f} \\ S_2 & \longleftarrow & S \end{array}$$

of affine schemes, with S_1, S_2 of finite type over $\mathbf{Z}$ and f_1, f_2 of finite type such that f_1 has property P and f_2 has property Q. Then we can find a cartesian diagram

$$\begin{array}{ccc} X_0 & \longleftarrow & X \\ {\scriptstyle f_0}\downarrow & & \downarrow{\scriptstyle f} \\ S_0 & \longleftarrow & S \end{array}$$

of affine schemes with S_0 of finite type over $\mathbf{Z}$ and f_0 of finite type such that f_0 has both property P and property Q.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The given pair of diagrams correspond to cocartesian diagrams of rings

$$\begin{array}{ccc} B_1 & \longrightarrow & B \\ \uparrow & & \uparrow \\ A_1 & \longrightarrow & A \end{array} \quad \text{and} \quad \begin{array}{ccc} B_2 & \longrightarrow & B \\ \uparrow & & \uparrow \\ A_2 & \longrightarrow & A \end{array}$$

Let $A_0 \subset A$ be a finite type $\mathbf{Z}$-subalgebra of A containing the image of both $A_1 \to A$ and $A_2 \to A$. Such a subalgebra exists because by assumption both A_1 and A_2 are of finite type over $\mathbf{Z}$. Note that the rings $B_{0,1} = B_1 \otimes_{A_1} A_0$ and $B_{0,2} = B_2 \otimes_{A_2} A_0$ are finite type A_0-algebras with the property that $B_{0,1} \otimes_{A_0} A \cong B \cong B_{0,2} \otimes_{A_0} A$ as A-algebras. As A is the directed colimit of its finite type A_0-subalgebras, by Limits, Lemma 01ZM we may assume after enlarging A_0 that there exists an isomorphism $B_{0,1} \cong B_{0,2}$ as A_0-algebras. Since properties P and Q are assumed stable under base change we conclude that setting $S_0 = \mathrm{Spec}(A_0)$ and

$$X_0 = X_1 \times_{S_1} S_0 = \mathrm{Spec}(B_{0,1}) \cong \mathrm{Spec}(B_{0,2}) = X_2 \times_{S_2} S_0$$

works. □